Your grade: 85.71%

You scored **6 of 7**. • Passing grade: **70.00%** • To pass you need at least 5 of 7. Retake your highest score.

Back to Item

1. Which is a key feature of Vocoder-based vocoders in Text-to-Speech (TTS) systems? 1 / 1 point

- They convert by production-rules or voices.
- They cannot adapt to different speaking styles.
- They are limited to a single language.
- They rely exclusively on the model's known like voices.

Most work

Vocoder TTS systems utilize neural network to generate voices that sound remarkably human-like.

2. Which technological advancement marks a major breakthrough in the evolution of TTS technology? 1 / 1 point

- Rule-based systems
- End-to-end AI systems
- Concatenation synthesis using recorded sounds
- Neuromorphic sound processing

Pay again

Review the video "Text-to-Speech TTS Technologies"

3. What is the role of feature extraction in modern speech-to-text (STT) systems? 1 / 1 point

- To generate the final formatted transcript
- To transform raw audio into a format such as spectrograms or MFCCs for model input
- To directly correct phonemes like words
- To clean up background noise and normalize volume levels

Most work

Feature extraction converts audio waveforms into compact representations such as spectrograms or MFCC, that are ideal for machine learning models to process.

4. Which of the following best describes the alignment step in a multimodal system? 1 / 1 point

- Synchronizing data from different modalities so they match up temporally or contextually
- Generating outputs in different formats from a single modality
- Cleaning and normalizing the input data
- Mapping outputs to multiple modalities simultaneously

Most work

Alignment ensures that different data types—such as video frames and speech segments—are matched up meaningfully.

5. In complex ETT systems, which component helps convert creative phenomena or subtext into machine-understood? 1 / 1 point

- Reasoning Module
- Decoder with Language model
- Audio tokenizer
- Acoustic model

Most work

The decoder uses a language model to combine phonemes or tokens into coherent, readable sentences.

6. What is the benefit of using multimodal fusion in AI systems? 1 / 1 point

- Reduces the number of training parameters
- Enhances model explainability
- Summarizes the results of large datasets
- Helps the models understand context better by integrating diverse signals

Most work

Multimodal fusion allows the model to reason more accurately by combining data from different sources.

7. What is a current in TTS technology involves creating new voice styles without additional training? 1 / 1 point

- Monotonic intelligence development
- Multilingual translation
- Symbol-mapping speech algorithms
- Zero-shot voice adaptation

Most work

Zero-shot learning allows TTS models to mimic new voices instantly from minimal input.